lift_scan x Feistel order: the combined n = 83 K600 one-target index calculus

2026-10-15

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Status (2026-10-15): measured; frozen primary rule applied; verdict `not_separated`. No new baseline. GitHub Actions run 38803625711 (attempt 1, one `ubuntu-24.04` runner, AMD EPYC 7763, Tier 1 instance scope) ran the frozen protocol and schedule. Its artifact `n83-combined-38803625711-1` is archived in `run_github_38803625711/`. Against the better single arm (lift_scan only), the combination is **1.236 [1.067, 1.428]** on means (R_m) and **1.061 [0.908, 1.243]** on geometric means (R_g). R_m clears 1; R_g does not, so the rule says `not_separated`. See Result.

Question

Two measured, promoted n = 83 a = 1 K600 one-target IC improvements change different factors of the online cost:

- **lift_scan** (PR #926, `claude/n83-liftscan` @ 8f6b4057e, run 38776966036) replaces the 83-rotation partner-root canonicalizer with the exact two-limb `lift_scan`. It cuts ns/probe by 1.515 [1.506, 1.525] and leaves the probes unchanged.
- **Feistel scan order** (PR #927, `claude/n83-scan-order` @ af41562e7, run 38786433493) visits the same index states in a keyed pseudo-random order instead of left-orbit-major. It cuts probes. On its held-out 384-point panel the ratio of means was 1.45 [1.20, 1.74], and mean probes over the law fell from 1.43 to 0.97.

Each was measured against B with the other factor at baseline. Does the combination beat the better of the two single changes on new targets? Under AGENTS.md's success criterion (PR #919), a verified, measured improvement over the current baseline is a success and becomes the new baseline.

The producer

`make_online_source.py` derives the producer from B's patched producer (crypto 770f6a69 plus B's `ic_log_cache.patch`). It applies the two parents' counted, exact string replacements in sequence:

1. lift_scan's edits, verbatim, plus one planted canonicalizer, `lift_plus_full`;
2. the scan order's edits, with the old strings adapted to the text after step 1, so the target scan carries both `<const C: u8, const O: u8>`.

Each process picks its arm with `KIC_TARGET_CANON` (`full`, `lift`, `lift_plus_full`) and `KIC_SCAN_ORDER` (`left_major`, `feistel`). The rank stage always uses the full scan and left-major order. Locally, the parents' own derivations reproduce their frozen producer digests from the same restored input, so both merges start from the exact measured text.

Frozen design

Arm	Canonicalizer	Order	Panel	Role
`pB`	full	left-major	primary	B
`pL`	lift	left-major	primary	lift_scan only

Arm	Canonicalizer	Order	Panel	Role
`pF`	full	Feistel	primary	Feistel only
`pLF`	lift	Feistel	primary	candidate (combined)
`pLFaa`	lift	Feistel	primary	A/A control
`pX`	lift + discarded full scan	Feistel	primary	planted positive control
`rT`	—	—	primary, every point	tuned rho rT770 (Fold/64/DP6), seed 840000 + point seed
`hB`, `hL`, `hF`, `hLF`	as above	as above	old 384 held-out	secondary
`oB`, `oL`, `oF`, `oLF`	as above	as above	Q0 + B's 16 fresh, 2 blocks	secondary

- **Primary panel.** 576 new public points (`inputs/primary_targets.json`, `make_targets.py`). They come from B's public hash-to-curve law with the new role `combined-primary-v1`, seeds 0..575. They are checked disjoint from Q0, the 16 fresh points and the 384 old held-out points. No scalar is known for any of them.
- **One job, one pinned CPU.** Nine primary slices of 64 points, interleaved with six old held-out slices and two Q0 + fresh blocks: P1, H1, P2, H2, P3, O1, P4, H3, P5, H4, P6, O2, P7, H5, P8, H6, P9. Within each segment the order is a seeded shuffle. Every primary slice holds one process per IC arm and one rho process per point.
- **Process model.** Each IC process does its setup once and then gives each target its own one-target online interval. No target-dependent work is shared between targets.
- **Primary statistic.** The better single arm over the combined arm:
  - R_m = min(mean `pL`, mean `pF`) / mean `pLF` (ratio of means);
  - R_g = min(gm `pL`, gm `pF`) / gm `pLF` (ratio of geometric means);
  - online time over all 576 points, bootstrap over points, 10,000 resamples. The better single arm is chosen again inside every resample.
- **Decision rule.**
  - **Improvement** iff both 95% lower bounds exceed 1. The combination then becomes the new n = 83 one-target baseline.
  - **Regression** iff both upper bounds are below 1.
  - Otherwise **not_separated**.
  - The verdict is **invalid** if any of these fails: an exactness gate, the A/A (`pLF`/`pLFaa` inside [0.97, 1.03]), the planted control (`pX`/`pLF` lower bound above 1), or the completeness of all 576 points.
- **Also reported.** Combined vs each single arm and vs B, probe ratios, the new `S_rho` and its delta, the tuned-rho gap on the same points, a slice-level bootstrap (sensitivity to process-level effects), and the old panels as secondaries.
- **Exactness gates.** On every primary point:
  - `pL` reproduces `pB`'s probes and relation;
  - `pF`, `pLF`, `pLFaa` and `pX` have identical probes and relation;
  - one scalar across all six IC arms and the tuned rho, replayed as [d]G = Q in the pinned oracle.

  On the old panels every arm must reproduce the probes and relation archived for its order (`inputs/archived_identities.json`), and Q0 must equal the validation sidecar.
- **Power, checked before freezing** (`power_check.py`, `analysis/power.json`).
  - The simulation resamples the old held-out panel's measured left-major and Feistel times, and the 16 measured full / lift_scan ratios.
  - R_m has power near 1 at every size tried. R_g is the binding statistic, with an expected 1.27 against 1.45 for R_m.
  - 576 points give 0.87 power for the conjunction. That is the smallest size tried whose power stays at least 0.80 after subtracting two Monte Carlo standard errors.
  - If the order effect were only 1.30 on means (the stress scenario), the power would be 0.39. That scenario is reported, not sized for.
- **Runtime estimate.** About 2.2 to 3.2 h, including the build. There is a 325 min time guard.

Local validation (cloud-agent VM; no timing claims)

The VM is a shared 4-vCPU guest, so its clocks are not evidence. See `smoke/` for the correctness-only receipt. In summary:

- Both crates build. The producer, rho and `lift2` tests pass, and B's factor-log table re-verifies.
- On primary points p000-p002, all six arms and the tuned rho are exact:
  - lift_scan matches the full scan's probes and relation;
  - the four Feistel arms are identical;
  - one scalar per point.
- On Q0 and the 16 fresh points, all four configurations reproduce the archived probes and relations of their order, and the archived scalars.

Files

File	Role
`protocol.json`, `make_protocol.py`, `schedule.json`, `make_schedule.py`	frozen protocol, its builder, frozen run order
`make_online_source.py`, `online/`, `lift2/`	combined producer (both parents' counted replacements), tuned rho, the two-limb canonicalizer crate
`restore.py`, `check_log_table.py`	restore and digest-check crypto 770f6a69 and every local source; re-verify B's factor-log table
`make_targets.py`, `inputs/primary_targets.json`, `inputs/old_heldout_targets.json`	the new primary panel; the old held-out panel, regenerated and compared
`inputs/archived_identities.json`	archived probes, relations and scalars per old point and order (exactness gates)
`power_check.py`, `inputs/power_prior.json`, `analysis/power.json`	power simulation and its prior
`arm.py`, `panels.py`, `run_schedule.py`	one arm per pinned process; panels; the schedule runner
`verify_receipt.py`	gates, statistics, decision rule; `result.json`, `result_table.md`
`run_github_38803625711/`	the archived artifact: allocation, builds, raw runs, `panel/result.json`
`make_figures.py`, `figures/`	result figure (TikZ source, PDF, PNG)
`README.pdf`	this report: `pandocREADME.md-oREADME.pdf--pdf-engine=pdflatex-Hreport_header.tex--lua-filterreport_filter.lua-Vgeometry:margin=2cm-Vfontsize=10pt-Vcolorlinks=true`

Result

All 662 scheduled rows are valid (576 primary points, 384 old held-out, 17 Q0 + fresh; no time-guard trip). Every exactness gate passes, the A/A and the planted control pass, and the primary panel is complete. `verify_receipt.py` re-run on the archive reproduces `result.json` byte for byte. One tuned-rho row (serial 5) carries a major-page-fault flag; it is a rho row, not an IC arm, and the verifier keeps it in the receipt.

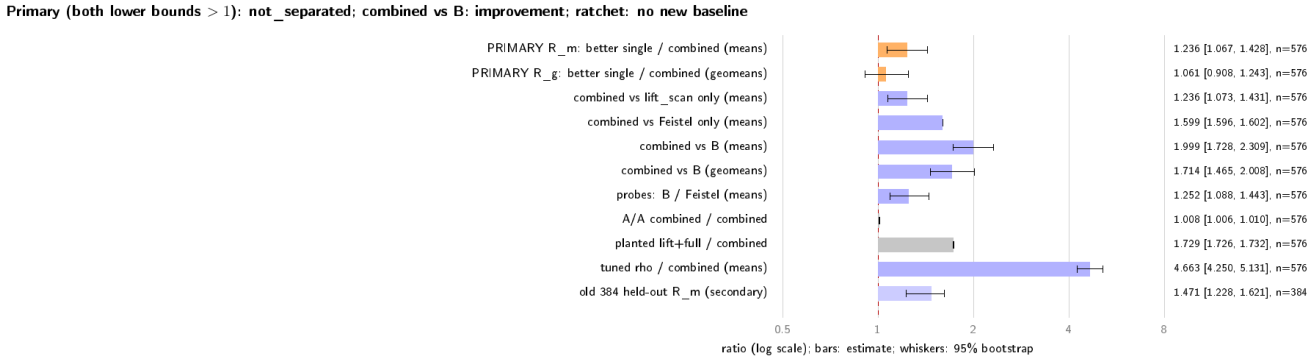


Figure 1: Frozen ratios with 95% bootstrap intervals, all from `result.json` (source, vector, rendered by `make_figures.py`).

Primary (frozen rule, applied literally)

The combination is an improvement only if both 95% lower bounds exceed 1.

Statistic (n = 576 new points)	Estimate [95%]	Reading
R_m = min(mean pL, mean pF) / mean pLF	1.2363 [1.0671, 1.4276]	lower bound above 1
R_g = min(gm pL, gm pF) / gm pLF	1.0611 [0.9083, 1.2430]	lower bound below 1

Verdict: `not_separated`. The R_g interval includes 1, so the frozen rule does not promote the combination. The better single arm is pL (lift_scan only) in both statistics' point estimates.

Secondaries (none changes the verdict)

- **Means-only reading.** R_m alone would read `improvement`. The protocol made the conjunction the rule, and R_g is the statistic the power check named as binding (expected 1.27; observed 1.06). The slice-level bootstrap of R_m (9 slices) gives 1.2363 [1.1005, 1.3957].
- **Combined against each single arm and against B** (ratio of means, then geometric means):

Contrast	Means [95%]	Geomeans [95%]
combined vs lift_scan only	1.2363 [1.0733, 1.4307]	1.0611 [0.9055, 1.2435]
combined vs Feistel only	1.5994 [1.5965, 1.6023]	1.5985 [1.5962, 1.6009]
combined vs B	1.9992 [1.7285, 2.3090]	1.7143 [1.4646, 2.0080]

lift_scan only vs B is 1.6171 [1.6156, 1.6186] on means; Feistel only vs B is 1.2500 [1.0825, 1.4408]. The probe ratio B / Feistel is 1.2523 [1.0877, 1.4430]. The two single effects multiply on means ($1.25 \times 1.62 = 2.02$, against 2.00 observed for the combination). On geometric means the order effect is small (Feistel vs B 1.0724 [0.9129, 1.2546]), which is why R_g is the binding statistic. We have not tested why; that is a reading, not a result.

- **Online means (ms).** pB 1072.8, pL 663.4, pF 858.3, pLF 536.6, pLFaa 532.3, pX 927.8. Probes per law: B and lift_scan 1.256, Feistel arms 1.003.
- **S_rho** (online cost over $\sqrt{r}$ in tuned-rho steps; the unit is this run's measured rT770 step): pB 0.04137, pL 0.02558, pF 0.03310, pLF 0.02069. Delta S_rho: combined minus better single (pL) = −0.00489; combined minus B = −0.02068. These are online S only; no cold S was measured.
- **Gap to tuned rho** (rT770 on the same 576 points). Tuned rho / combined is 4.6628 [4.2501, 5.1308] on means and 7.2786 [6.5115, 8.1684] on geometric means; the same figure for B is 2.3323 [2.0713, 2.6339]. So the online interval of the combination is about 0.214 of tuned rho's on means (inverse of 4.66). This is distance and not a gate; it is online only, and the precomputation is not in it.

- **Controls.** A/A pLF / pLFaa = 1.0081 [1.0058, 1.0104], inside the frozen [0.97, 1.03]. Planted pX / pLF = 1.7290 [1.7256, 1.7321], lower bound above
   1.
- **Old panels (secondary).** Held-out 384: R_m 1.4710 [1.2283, 1.6213] (`improvement`), R_g 1.2942 [1.0757, 1.5564], combined vs B 2.3876 [1.9776, 2.8539]. Q0 + 16 fresh (one block of 16): R_m 1.6237 [0.9876, 1.6270], combined vs B 5.5007 [1.6284, 11.9937]; n = 16 is too small to read.

Reading

- The primary rule is not met. The combination is clearly faster than B (about 2.0× on means, 1.7× on geometric means), and it beats Feistel alone by 1.60× on both. Against lift_scan alone it separates on means but not on geometric means.
- The held-out panel, a separate and larger-effect sample, does separate on both statistics, so the not-separated verdict is about this panel's R_g, not evidence of no effect. It is a secondary and does not change the rule.
- Ratchet: no new baseline from this experiment. See `docs/ic/BASELINES.md`.

Process note

A collector stopped by a container restart was replaced; nothing was archived before this result, and no run was repeated.

Limits

- Tier 1, instance-scoped: one runner, one pinned CPU, not host-isolated.
- Each IC process serves a 64-point slice warm, with one online interval per target and no shared target-dependent work. Process-level speed shifts are shared within a slice, which is why the slice bootstrap is reported.
- The tuned-rho gap is distance remaining. It is not a claim to beat rho.